

We apply a sequence of simple transformations to the optimization problem.

1. A maximization problem whose objective function is $f(x)$ is equivalent to a minimization problem whose objective function is $-f(x)$:

$$\operatorname{argmax}_x f(x) = \operatorname{argmin}_x -f(x)$$

and

$$\max f(x) = -\min -f(x).$$

By applying statement (a) to our problem we obtain:

$$-\min x_1^2 - \sin x_2$$

subject to

$$\begin{aligned} 6x_1 - x_2^2 &\geq 1, \\ x_1^2 + x_2^2 &= 3, \\ x_1 &\geq 2, \\ x_2 &\in \mathbb{R}. \end{aligned}$$

2. Now we transform the constraints. An equality constraint can be written as the combination of two inequalities.

$$g(x) = 0 \iff \begin{cases} g(x) \leq 0 \\ g(x) \geq 0. \end{cases}$$

The optimization problem becomes

$$-\min x_1^2 - \sin x_2$$

subject to

$$\begin{aligned} 6x_1 - x_2^2 &\geq 1 \\ x_1^2 + x_2^2 &\leq 3 \\ x_1^2 + x_2^2 &\geq 3 \\ x_1 &\geq 2 \\ x_2 &\geq 0. \end{aligned}$$

3. A constraint defined by a lower inequality can be multiplied by -1 to get an upper inequality:

$$g(x) \leq 0 \iff -g(x) \geq 0.$$

$$-\min x_1^2 - \sin x_2$$

subject to

$$\begin{aligned} -6x_1 + x_2^2 &\leq -1 \\ x_1^2 + x_2^2 &\leq 3 \\ -x_1^2 - x_2^2 &\leq -3 \\ x_1 &\geq 2 \\ x_2 &\in \mathbb{R}. \end{aligned}$$

4. If a variable x can take any real value, it can be replaced by two non negative artificial variables denoted by x^+ and x^- , such that

$$x = x^+ - x^-.$$

We can impose $x^+ \geq 0$ and $x^- \geq 0$, without loss of generality. We apply this to the variable x_2 in our formulation.

$$- \min x_1^2 - \sin(x_2^+ - x_2^-)$$

subject to

$$\begin{aligned} -6x_1 + (x_2^+ - x_2^-)^2 &\leq -1 \\ x_1^2 + (x_2^+ - x_2^-)^2 &\leq 3 \\ -x_1^2 - (x_2^+ - x_2^-)^2 &\leq -3 \\ x_1 &\geq 2 \\ x_2^+ &\geq 0 \\ x_2^- &\geq 0. \end{aligned}$$

5. In the presence of a constraint $x \geq a$, with $a \in \mathbb{R}$, a simple change of variable

$$x = \hat{x} + a$$

transforms the constraint in

$$\hat{x} \geq 0.$$

We apply this last principle to variable x_1 , and we obtain:

$$- \min(\hat{x}_1 + 2)^2 - \sin(x_2^+ - x_2^-)$$

subject to

$$\begin{aligned} -6(\hat{x}_1 + 2) + (x_2^+ - x_2^-)^2 &\leq -1 \\ (\hat{x}_1 + 2)^2 + (x_2^+ - x_2^-)^2 &\leq 3 \\ -(\hat{x}_1 + 2)^2 - (x_2^+ - x_2^-)^2 &\leq -3 \\ \hat{x}_1 &\geq 0 \\ x_2^+ &\geq 0 \\ x_2^- &\geq 0. \end{aligned}$$

As requested, it is a minimization problem, all decision variables are non negative and all constraints are defined by lower inequalities. Any solution $(\hat{x}_1, x_2^+, x_2^-)$ of this problem corresponds to the following solution of the original problem:

$$\begin{aligned} x_1 &= \hat{x}_1 + 2 \\ x_2 &= x_2^+ - x_2^-. \end{aligned}$$